RH-Conditional Approximation Estimates and the Remaining Simplicity Problem

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Research consolidation draft

Abstract

We consolidate analytic estimates obtained while investigating whether the Riemann hypothesis implies simplicity of the nontrivial zeros of the zeta function. The principal approximation argument establishes, within the research notes, a complete first-order limit for fixed Gevrey-tapered Möbius polynomials and an RH-conditional infimum constant equal to the sum of squared multiplicities. A completed-ratio energy has an explicit heat-scale transition governed by the same multiplicity excess. Exact finite Dirichlet filters give a sharp local resolution cost. These estimates do not exclude a multiple zero: the missing ingredient is an arithmetic upper estimate that contradicts its additional cost. This brief presents precise statements and proof mechanisms, with the full analytic payments referenced to the research notes.

1 Scope and the unresolved quantity

Throughout, RH is an assumption, not a conclusion. Write $\rho = \frac{1}{2} + i\gamma$ for a distinct nontrivial zero, m_ρ for its multiplicity, and sum over both ordinate signs. The two masses are

$$K = \sum_{\rho \text{ distinct}} \frac{m_\rho^2}{|\rho|^2}, \qquad Z = \sum_{\rho \text{ distinct}} \frac{m_\rho}{|\rho|^2} = 2 + \gamma_E - \log(4\pi). \tag{1}$$

They are finite; the usual unit-window zero count also bounds the sum of squared multiplicities with this weight. Their difference is

$$K - Z = \sum_{\rho \text{ distinct}} \frac{m_\rho(m_\rho - 1)}{|\rho|^2}. \tag{2}$$

A double conjugate pair contributes $4/(\gamma^2 + 1/4)$. The approximation and heat estimates below retain this excess. Identifying it is a bookkeeping conclusion, not a new simplicity criterion or a proof that the excess vanishes.

The results inherited from boundary winding justify the local multiplicity calculation and finite-height continuation under RH. They do not furnish a strict surplus with the opposing sign. The present judgment is therefore that the investigation has produced analytic estimates, but does not have a nearly complete simplicity proof.

Status. The new analytic results are proved in the linked research notes and have undergone separate same-model internal derivations and adversarial checks. They have not received specialist external review or analytic formalization. In particular, the complete scaled-tail theorem below is a substantial internal premise, not a previously published theorem being cited under a new name. No novelty or priority claim is made.

2 Fixed Gevrey approximation and the complete tail

For a Dirichlet polynomial D, define

$$E(D) = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{|1-\zeta(\frac{1}{2}+it)D(\frac{1}{2}+it)|^2}{1/4+t^2}\,\mathrm{d}t, \qquad d_N^2 = \inf_{D(s)=\sum_{n\le N} a_n n^{-s}} E(D).$$

Let W be real, smooth, supported in $[-1,1]$, with $W(0)=1$ and

$$\|W^{(j)}\|_1 \le A^{j+1}(j!)^\sigma, \qquad 1<\sigma<2.$$

Set $w = W|_{[0,1]}$, $L=\log N$, and

$$V_{N,w}(s) = \sum_{n\le N} \mu(n) n^{-s} w(\log n/L), \qquad p_{m,w}(u) = -\frac{(u^{m-1}w(u))^{(m)}}{(m-1)!}, \qquad C_m(w) = \|p_{m,w}\|^2_{L^2(0,1)}.$$

Theorem 1 (Complete fixed-taper estimate)**.** *Assume RH and fix W as above. If $E_{N,w}^{>T}$ denotes the part of $E(V_{N,w})$ at $|t|>T$, then for every $0<\eta<2-\sigma$,*

$$\sup_{N\ge 2} (\log N) E_{N,w}^{>T} \ll_{W,\eta} T^{\sigma-2+\eta}, \qquad T\ge 2, \tag{3}$$

$$\lim_{N\to\infty} (\log N) E(V_{N,w}) = \sum_{\rho \text{ distinct}} \frac{C_{m_\rho}(w)}{|\rho|^2} < \infty. \tag{4}$$

No simplicity, zero-gap lower bound, or inverse-derivative moment is assumed.

Proof mechanism and dependencies. The full proof is Note 16, Sections 2–6, 8 and 11. For $s=\frac{1}{2}+it$ and $H>0$, its exact starting identity is

$$V_{N,w}(s) = \int_{\mathbb{R}} \frac{K_H(v)}{\zeta(s+(H+iv)/L)}\,\mathrm{d}v, \quad K_H(v) = \frac{1}{2\pi}\int_{\mathbb{R}} W(u) e^{(H+iv)u}\,\mathrm{d}u.$$

Fourier inversion on an absolutely convergent Möbius line, followed by a shift within the RH zero-free half-plane, preserves every coefficient. Integration by parts against the whole complex exponential gives

$$|K_H(v)| \le C_W e^H e^{-c_W|v|^{1/\sigma}}, \qquad \int_{\mathbb{R}} K_H(v)\,\mathrm{d}v = 1,$$

with constants independent of $H>0$. The factor e^H is retained.

The primary height inputs are the RH argument bound of Carneiro–Chandee–Milinovich [2] and the subpower zeta bound of Chandee–Soundararajan [3]. In the low-height range, the proof keeps the whole nearby zero cluster, including repetitions. Its kernel moment has logarithm $(\sigma-1)M\log(M+2)+O_W(M+1)$ for a cluster of size M. This is paid using the half-coefficient local counting allowance; no individual reciprocal derivative is estimated. A separate moving line handles the high-height range and its complete remote convolution.

Subtracting the constant term coherently gives decay in the actual distance to the nearest zero. Squaring and integrating that decay over nearest-zero cells costs $O(1/L)$ independently of the gaps. This proves (3). On a fixed isolated zero neighborhood, the complete error has profile $\int_0^1 p_{m,w}(u)e^{-iLyu}\,\mathrm{d}u + O_{W,\rho}(L^{-1})$, with constants allowed to depend on that neighborhood. Its unscaled norm is $C_m(w)/(L|\rho|^2)+O(\log L/L^2)$. The separately established tail bound then permits height exhaustion. Thus local asymptotics alone are not being used as a global upper bound.

A controlled moving family. For the almost-linear taper $w_\eta(u) = (1-u)\chi((1-u)/\eta)$, with a fixed Gevrey cutoff $\chi = 0$ on $(-\infty, \frac{1}{2}]$ and $\chi = 1$ on $[1, \infty)$, a separate proof gives

$$E(V_{N,w_\eta}) \le \frac{C\exp(C\eta^{-a})}{L}, \quad L^{-\theta_0} \le \eta < \tfrac{1}{4}, \quad a > \frac{1}{2-\sigma}, \quad 0 < \theta_0 < 1. \tag{5}$$

Here the parameters σ, θ_0, a are fixed (Note 16, Section 12). Consequently $\eta_N = (\log L)^{-\kappa}$, $0 < \kappa < 2-\sigma$, gives $E(V_{N,w_{\eta_N}}) \le L^{-1+o(1)}$. The exponential allowance is compatible with the multiple-zero lower costs. It supplies neither a moving first-order asymptotic nor a multiplicity-free upper constant.

3 The infimum constant and finite Dirichlet filters

Integration by parts gives $\int_0^1 u^j p_{m,w}(u)\,\mathrm{d}u = \delta_{j0}$ for $0 \le j < m$. Projection onto polynomials of degree below m therefore yields

$$C_m(w) = m^2 + \|p_{m,w} - \mathcal{K}_m(\,\cdot\,, 0)\|_2^2 \ge m^2, \tag{6}$$

where $\mathcal{K}_m$ is their endpoint reproducing kernel on $[0,1]$. Flat Gevrey approximations to $(1-u)^m$ approach this lower cost. One common taper need not optimize different multiplicities at once.

Theorem 2 (Infimum consequence of Theorem 1)**.** *Assuming RH and the complete fixed-taper estimate,*

$$\lim_{N\to\infty} (\log N) d_N^2 = K. \tag{7}$$

Real coefficients suffice.

Proof outline. Burnol's lower bound [1] gives the lower limit K. For the upper bound, fix a baseline taper and a finite set F of zeros carrying all but a small part of its profile mass. Choose a fixed near-optimal taper separately at each target. Bounded finite Dirichlet multipliers P_ρ select these profiles:

$$D_N = V_{n,w_0} + \sum_{\rho\in F} P_\rho(V_{n,w_\rho} - V_{n,w_0}), \qquad n = \lfloor N/M \rfloor,$$

where all multipliers have length at most M. The multiplier coefficients of the error sum identically to one, so every cross term and moment constraint is retained. For fixed multipliers, the local profile expansion together with Theorem 1's scaled-tail bound gives the complete mixed limit. Its profile tail is bounded by a fixed multiple of the sum of the finitely many original profile tails.

One may interpolate exactly on any larger finite zero set G. For $0 < \theta < \pi$, $k = \lfloor \pi/\theta \rfloor$, the polynomial

$$B_\theta(z) = \frac{\sin\theta - \sin((k+1)\theta)z^k + \sin(k\theta)z^{k+1}}{\sin\theta - \sin((k+1)\theta) + \sin(k\theta)}$$

has nonnegative coefficients summing to one, equals one at 1, and vanishes at $e^{\pm i\theta}$. At $\theta = \pi$, use $(1+z)/2$. Substitute $z = p^{\rho-s}$, choosing $p = 2$ or 3 with nonzero relative phase, and multiply over the other nodes of G. Irrationality of $\log 2/\log 3$ guarantees such a choice. The products remain bounded by one on the critical line. Their complete exterior profile allowance is independent of their degrees and tends to zero as G exhausts the zeros.

Keep F, tapers and finite G fixed before taking $N \to \infty$. Then exhaust G, optimize the finite profile costs, and let the initial baseline-tail tolerance tend to zero. Conjugate pairing makes the final coefficients real. This proves the upper limit K; see Note 16, Sections 13–14. □

This conclusion concerns the unrestricted polynomial infimum, not the original linear Möbius polynomial. It removes a common-taper restriction but leaves the squared multiplicities in (1). It provides no effective threshold in N.

Proposition 3 (Sharp two-point isolation cost)**.** *Let P have Dirichlet length at most M, be bounded by one on $\Re s = \frac{1}{2}$, and satisfy $P(\rho) = 1$ and $|P(\rho + i\Delta)| \le \varepsilon$, $0 \le \varepsilon < 1$. Then*

$$|\Delta| \log M \ge 2 \arccos \varepsilon.$$

For exact cancellation at a small nonzero gap, the least log-length lies between $\pi/|\Delta|$ and $\pi/|\Delta| + \log 2$.

Proof. Centering the frequency interval gives an entire function of type at most $(\log M)/2$. Its real part starts at one, is bounded by one, and ends at most ε. The sharpened Bernstein inequality [4], integrated after arccos, gives the lower bound. The polynomial B_θ with $p = 2$ and $\theta = |\Delta| \log 2$ gives the upper bound. □

This prices root isolation only. Assigning the same taper to two roots need not isolate them, and the proposition is not a lower bound for every approximation scheme.

4 Completed-ratio energy and heat separation

Use the original theta source

$$H(t,x) = \int_0^\infty e^{tu^2} \Phi(u) \cos(xu)\, \mathrm{d}u, \qquad \Phi(u) = \sum_{n \ge 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}},$$

with $\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s)$. The completion and heat normalization are

$$\mathcal{F}_t(z) = 8H(t, -2iz), \quad \mathcal{F}_0(z) = \xi(\tfrac{1}{2} + z), \quad \partial_t \mathcal{F}_t = \tfrac{1}{4} \partial_z^2 \mathcal{F}_t.$$

For $\alpha > 0$, define

$$J_t(\epsilon) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|1 - \mathcal{F}_t(iy)/\mathcal{F}_t(\epsilon + iy)|^2}{\alpha^2 + y^2}\, \mathrm{d}y, \qquad M_\alpha = \sum_\gamma \frac{m_\gamma}{\alpha^2 + \gamma^2}, \quad K_\alpha = \sum_\gamma \frac{m_\gamma^2}{\alpha^2 + \gamma^2}.$$

Under RH, for $t \ge 0$, the paired order-one product makes the ratio contractive. Its positive harmonic majorant has a known complete mass. Local profiles followed by exhaustion of that mass give

$$\lim_{\epsilon \downarrow 0} \frac{J_0(\epsilon)}{\epsilon} = \sum_\gamma \frac{c_{m_\gamma}}{\alpha^2 + \gamma^2}, \qquad c_m = m \left(1 - \frac{\binom{2m}{m}}{4^m} \right). \tag{8}$$

In particular $c_1 = \frac{1}{2}$, $c_2 = \frac{5}{4}$; no mass at infinity is discarded. The proof is Note 15, Sections 1–3.

For $t = v\epsilon^2$, the local heat expansion at an order-m zero gives Hermite centers $r\lambda_j$, where $r = \sqrt{v/2}$ and $\mathrm{He}_m(\lambda_j) = 0$. Define

$$P_{m,r}(y) = \prod_{j=1}^m (y - r\lambda_j), \qquad \mathcal{C}_m(r) = \frac{1}{2\pi} \int_{\mathbb{R}} \left| 1 - \frac{P_{m,r}(y)}{P_{m,r}(y - i)} \right|^2 \mathrm{d}y.$$

The same complete-mass argument proves, at each fixed $v \ge 0$,

$$\lim_{\epsilon \downarrow 0} \frac{J_{v\epsilon^2}(\epsilon)}{\epsilon} = \mathcal{A}_\alpha(v) := \sum_\gamma \frac{\mathcal{C}_{m_\gamma}(\sqrt{v/2})}{\alpha^2 + \gamma^2}.$$

The analytic transition is quantified by

$$\mathcal{C}_2(r) = 1 + \frac{1}{4(1+r^2)}, \tag{9}$$

$$0 \le \mathcal{A}_\alpha(v) - \frac{M_\alpha}{2} \le \frac{K_\alpha - M_\alpha}{4v}, \qquad v > 0, \tag{10}$$

$$\lim_{v\to\infty} v\left(\mathcal{A}_\alpha(v) - \frac{M_\alpha}{2}\right) = \frac{K_\alpha - M_\alpha}{4}. \tag{11}$$

The estimate follows from pairwise overlap integrals of the local contractive factors and $\sum_{i<j}(\lambda_i - \lambda_j)^{-2} = m(m-1)/8$. The weighted multiplicity-square sum dominates the passage to all zeros. At $\alpha = \frac{1}{2}$, the last coefficient is $(K - Z)/4$.

These are ordered limits: first $\epsilon \to 0$ at fixed v, then $v \to \infty$. They do not assert convergence of time derivatives or a joint limit with unbounded $v(\epsilon)$. The heat-deformed completion has not been assigned an Euler product or the original zeta Dirichlet convolution.

5 Relation to the expanding-contour RH investigation

The originating BW09 route seeks to rule out a hypothetical positive all-real threshold by forcing a later collision. Its current manuscript assumes an attained finite-height ordinary event at that threshold; coverage of every hypothetical positive threshold remains a separate unproved step. BW10 assumes RH and starts from a multiple zero at the specified time zero, allowing arbitrary finite multiplicity. It removes that unknown-time and attainment issue from its conditional problem. This is a structural distinction, not a proved ordering of difficulty.

The unchanged theta source, local Hermite splitting, root-wall cancellation and complete boundary-phase identities are shared. The ratio-energy analysis also applies relative to a known all-real base time T, with source $e^{Tu^2}\Phi(u)$, provided the all-real forward continuation and order-one product hypotheses are retained. The finite-selector estimates are unconditional filter lemmas. These tools sharpen the accounting of an existing event; they do not supply additional winding.

The strongest approximation estimates have a narrower scope. Theorem 1 uses RH for the original zeta function and its unchanged Möbius convolution. Importing that assumption into BW09's positive-threshold contradiction would assume the desired conclusion. All-realness of a positive-time heat deformation supplies neither this original Dirichlet arithmetic nor a substitute theorem.

BW09 still needs a complete theta estimate forcing an ordinary born pair's cap-phase decrease strictly beyond π, with wall continuation, a regular terminal cap, endpoint deficits and source remainders all controlled. Its fixed-index large-time cosine model produces the ordinary π, not a surplus. BW10 has not supplied that strict estimate or the threshold-event coverage step. See the BW09 manuscript, Sections on scope, root walls, surplus and information limits, and its phase-surplus progress register.

6 What is missing, and what to brainstorm next

The common bottleneck is now explicit. The complete approximation upper bound has coefficient K, and the heat transition has excess coefficient $(K - Z)/4$. Both are compatible with a multiple endpoint. A simplicity proof needs an additional estimate using the unchanged arithmetic of zeta that rules out this excess, or a different mechanism that excludes an original multiple zero.

The tested absolute prime and Möbius-correlation payments do not supply that estimate. In the exact ratio-energy representation, signed off-diagonal terms cancel a diagonal of size $1/\epsilon$ to bounded size, with the residual energy determined at absolute precision $O(\epsilon)$; taking absolute values gives a divergent double payment (Note 15, Section 6). The prime-error linearization also has a first-order remainder at multiple zeros (Section 7). Neither remainder can be omitted at the scale where simplicity is being tested. Generic positive-source collision controls demonstrate the insufficiency of source positivity alone, while failing the ordinary Dirichlet-series premise; they are not counterexamples satisfying all of zeta's arithmetic properties.

The next discussion should focus on two questions:

1. Is there a specific signed arithmetic identity whose *complete* remainder can be bounded more sharply than these multiplicity-compatible estimates? A proposed bound must retain all zero and prime tails.
2. Is there a theta-specific sign or rigidity principle, absent from the screened collision controls, that excludes the multiple endpoint without merely rewriting the desired conclusion?

Before further expansion, the complete scaled-tail theorem deserves external scrutiny because several later conclusions depend on it. The present estimates warrant consolidation and a change of viewpoint, not a claim of proximity to a proof.

Supporting checks. The repository's exact-algebra replay and bounded Lean check support selected algebraic identities and a bounded inner-product tradeoff. Numerical investigations in the originating boundary-winding manuscript provide context. None verifies the analytic tail theorem or simplicity; no numerical evidence is used as a proof premise here.

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References

[1] J.-F. Burnol, *A lower bound in an approximation problem involving the zeros of the Riemann zeta function*, Theorem 1.3. arXiv:math/0103058.

[2] E. Carneiro, V. Chandee and M. B. Milinovich, *Bounding $S(t)$ and $S_1(t)$ on the Riemann hypothesis*, Theorem 2. arXiv:1309.1526.

[3] V. Chandee and K. Soundararajan, *Bounding $|\zeta(1/2+it)|$ on the Riemann hypothesis*, Theorem 1.1. arXiv:0908.2008.

[4] R. M. Redheffer, *On a theorem of Plancherel and Pólya*, Pacific J. Math. **3** (1953), 823–835, Section 5, Theorem 11. Primary text.